



Solar Power 101

From Sun to Socket

The Complete Beginner-to-Pro Photovoltaic Guide

PV Arrays • Charge Controllers • Batteries • Inverters • Loads

Awet G. Nway

Robotics, Automation & MEP Engineering Specialist

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Welcome to Solar Power 101

Solar electricity is no longer an experimental curiosity — it is the fastest-growing energy source on Earth and the most accessible way to electrify rural communities, industrial sites, and modern homes alike. This guide walks you through the complete photovoltaic (PV) chain — from the photons leaving the sun to the alternating current powering your fridge.

Each module is built around a working schematic, real component data, and the kind of design rules-of-thumb engineers use on actual installs. By the end you will be able to size a small off-grid system, choose between PWM and MPPT controllers, and explain why an inverter's waveform matters for sensitive electronics.

What you will learn

Module	Topic	Key takeaway
1	Solar Array — Capturing the Sun	How a PV cell turns photons into DC current.
2	Charge Controller — The Power Manager	Why MPPT recovers 15-30% more energy than PWM.
3	Battery — Storing Energy for Later	Depth-of-discharge, chemistry trade-offs & cycle life.
4	Inverter — Converting DC to AC	Pure sine vs modified sine for sensitive loads.
5	The Load — DC vs AC consumption	Topology choice: off-grid, grid-tied, or hybrid.

i **INFO** Who this guide is for

Engineering students, technicians, and small-business owners planning a first solar install. Anyone comfortable with basic electricity (volts, amps, watts) can follow along — the math is kept practical, with worked examples at every stage.

0 The Big Picture: Sun to Socket

Before diving into individual components, study the complete chain. Every off-grid PV system routes energy through the same five stages — only the size and quality of each stage change between a 10-watt garden light and a 10-kilowatt rural microgrid.

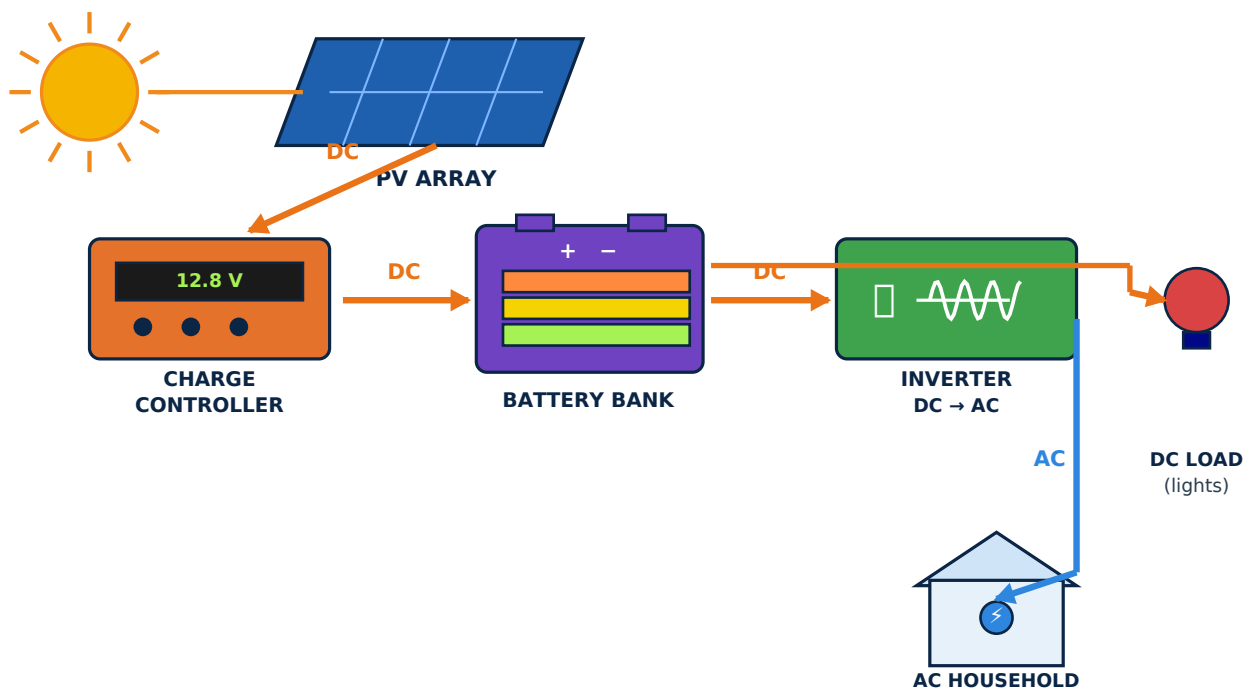


Figure 0.1 — Complete sun-to-socket energy flow with DC (orange) and AC (blue) sections

✓ TIP Read the wire colour, read the system

Throughout this guide, orange lines = DC and blue lines = AC. This is also the convention used on most solar wiring diagrams and in cable colour codes for battery-bank wiring. Memorising it makes troubleshooting much faster.

1 Solar Array — Capturing the Sun

A solar panel — properly called a photovoltaic (PV) module — is a collection of silicon cells wired in series. Each cell behaves like a tiny diode that produces a voltage of roughly 0.5 V whenever sunlight knocks electrons loose from the silicon crystal lattice. Sixty cells in series → about 30 V open circuit, which is the classic "12 V battery-charging" panel size.

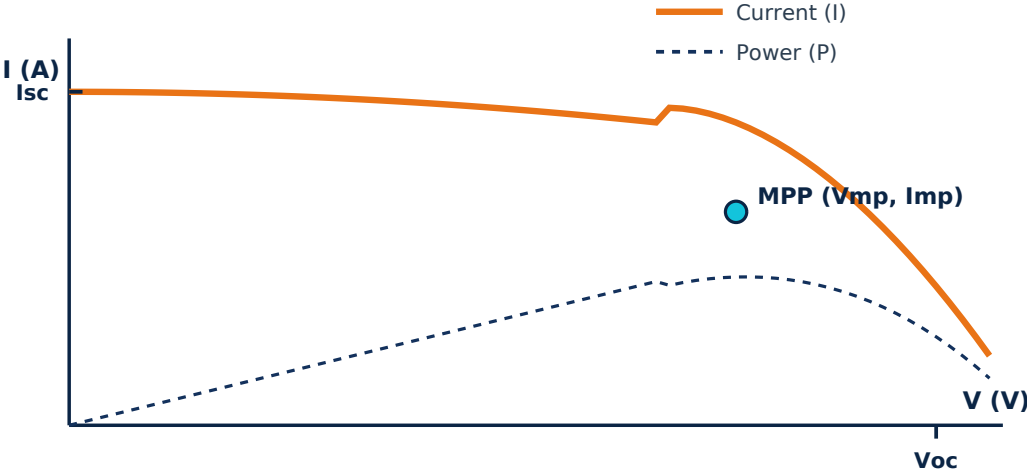


Figure 1.1 — Typical PV module I-V curve with Maximum Power Point (MPP)

Key electrical parameters

Parameter	Symbol	What it tells you
Open-circuit V	Voc	Voltage with no load — sets controller's max input.
Short-circuit I	Isc	Current into a dead short — used for fuse sizing.
Voltage at MPP	Vmp	Operating voltage at peak power; ~80% of Voc.
Current at MPP	Imp	Operating current at peak power; ~93% of Isc.
Peak power	Pmax	Rated wattage at STC (1000 W/m², 25 °C, AM 1.5).
Temp coeff.	β(Voc)	Voltage rises in cold; size strings for −10 °C site.

✓ TIP Quick rule

Real-world output ≈ 75–80% of nameplate Pmax after losses from temperature, wiring, soiling, and inverter efficiency. A 400 W panel delivers about 300–320 W average on a clear midday hour.

2

Charge Controller — The Power Manager

A charge controller is the regulator between the array and the battery. Without it, a fully-charged battery on a sunny day would be cooked by overvoltage in hours. The controller throttles the current and clamps the voltage to a safe charging profile — bulk (constant current), absorption (taper), and float (trickle).

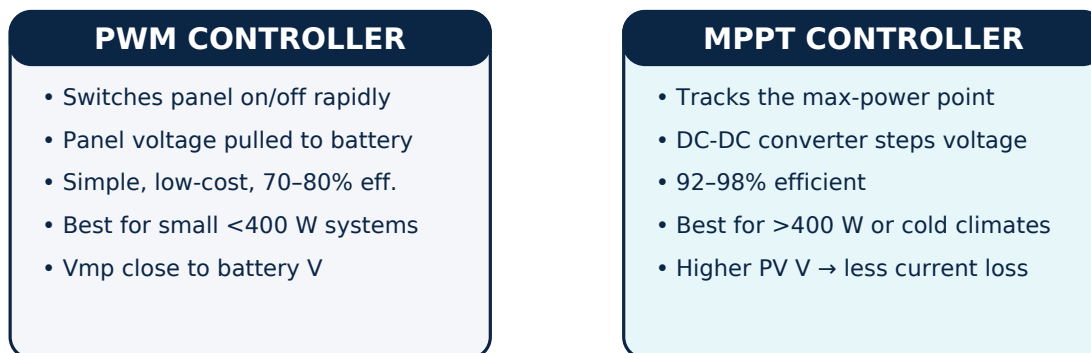


Figure 2.1 — PWM versus MPPT charge controller capabilities

Sizing rule for the controller

$I_{max} = P_{max_array} \div V_{battery} \times 1.25$ — multiply by 1.25 (NEC safety factor) and round up to the next commercial controller rating.

Example: A 400 W array charging a 12 V bank needs at minimum $(400 \div 12) \times 1.25 = 41.7$ A → choose a 50 A MPPT controller.

i INFO When to pick MPPT over PWM

Pick MPPT whenever (a) your panel Vmp is much higher than the battery voltage (e.g. 60-cell modules on a 12 V bank), (b) the array exceeds about 400 W, or (c) the site is cold — where rising panel voltage gives MPPT a bigger advantage.

3 Battery — Storing Energy for Later

Solar panels only produce power when the sun shines, but most loads need power 24 hours a day. The battery bank is the bridge — it stores midday surplus as chemical energy and releases it at night, during cloud cover, or under sudden load surges.

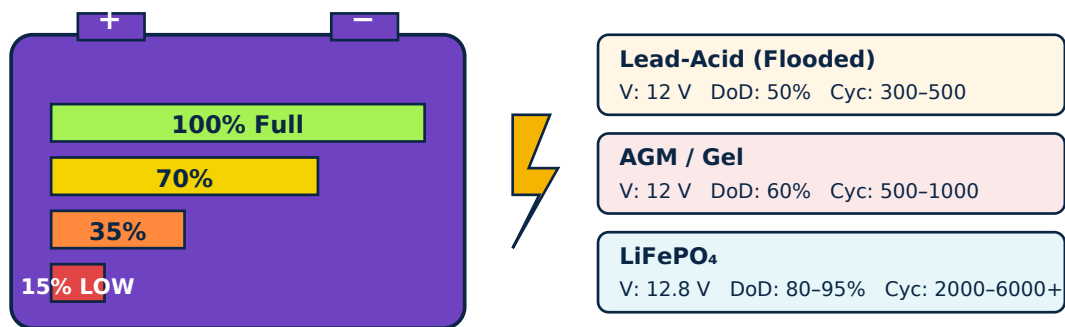


Figure 3.1 — State of charge bands and common battery chemistries

Sizing the battery bank

$$\text{Bank Wh} = \text{Daily load (Wh)} \times \text{Days of autonomy} \div \text{DoD} \div \eta_{\text{inv}}$$

Example: 1500 Wh/day load, 2 days autonomy, 80% DoD (LiFePO₄), 92% inverter → $1500 \times 2 \div 0.80 \div 0.92 \approx 4080$ Wh. At 12.8 V that's 320 Ah of lithium (or roughly 640 Ah of lead-acid at 50% DoD).

! WARNING Never deep-discharge lead-acid below 50%

Each discharge below 50% DoD on a flooded lead-acid battery removes hundreds of cycles from its life. LiFePO₄ has no such penalty down to 80% DoD — which is why it now dominates new off-grid installs despite the higher upfront cost.

4 Inverter — Converting DC to AC

Batteries store DC, but the wall sockets in your house run on AC at 50 Hz (or 60 Hz in the Americas). The inverter performs that conversion, electronically chopping the DC into a waveform that approximates a sine wave. The quality of that approximation matters more than most users realise.

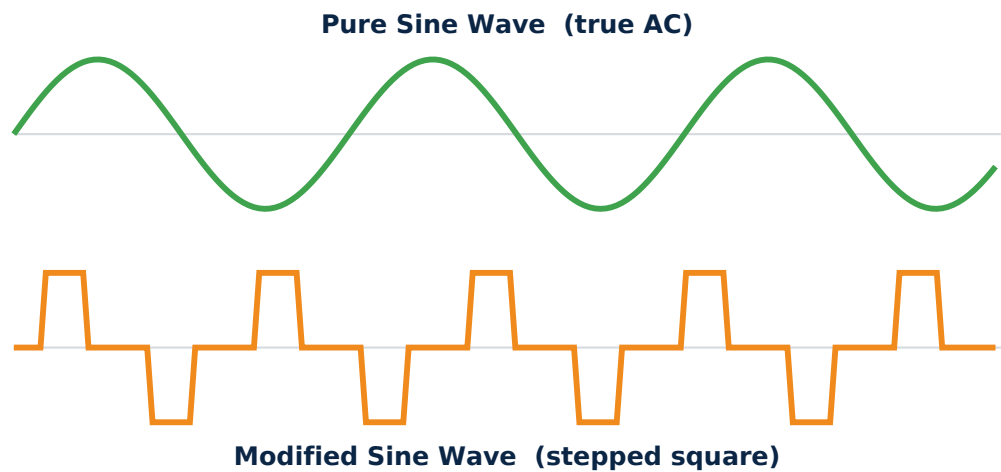


Figure 4.1 — Pure sine wave (top) versus modified sine wave (bottom)

Topic	Pure Sine Inverter	Modified Sine Inverter
Waveform	Smooth sinusoid identical to grid	Stepped square ('chopped') approximation
Cost	Higher (2-4× the modified type)	Low — cheap to manufacture
Motors / pumps	Run cool, full torque, full life	Run hot, lose ~20% torque, age fast
Electronics	Safe for laptops, medical, audio	Buzzing in audio; risk to sensitive gear
Best use	Permanent homes, sensitive loads	Lights and resistive loads only

✓ TIP Inverter sizing

Continuous rating must cover the steady load, but surge rating must cover the starting current of motors (compressors, pumps, fridges), which can briefly hit 5-7× their running watts. A 1500 W fridge compressor wants a 3000 W surge-capable inverter, even if its label says 250 W.

5 The Load — DC, AC & System Topology

The load is whatever consumes the energy. Some loads — LED lights, USB ports, 12 V refrigerators, and DC water pumps — run directly on battery voltage and need no inverter, saving 5-8% in conversion losses. Most household appliances, however, expect AC and force the inverter into the chain.

Three system topologies

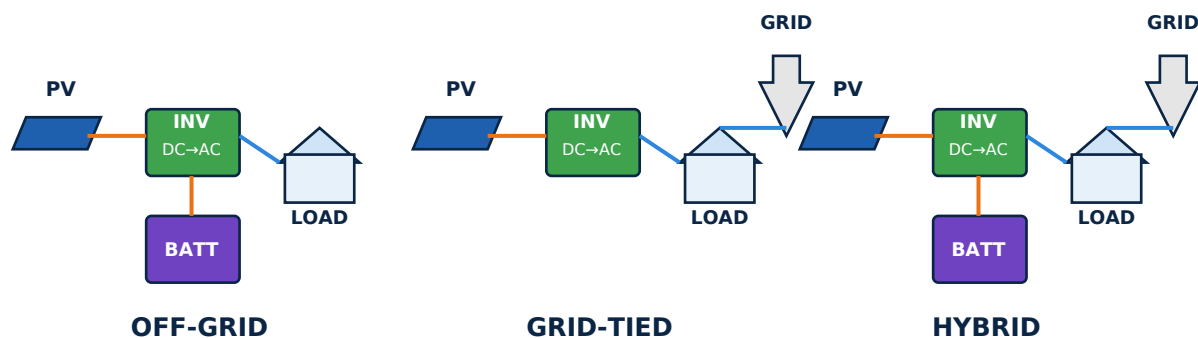


Figure 5.1 — Off-grid, grid-tied, and hybrid PV system topologies

Topology	Battery	Grid	Best for
Off-grid	Yes	No	Rural homes, mobile / marine, no utility access.
Grid-tied	No	Yes	Urban bill reduction; cheapest €/W; no backup.
Hybrid	Yes	Yes	Critical loads + grid backup + bill savings.

× **DANGER** Never back-feed the grid without an anti-islanding inverter

If utility power fails and your solar inverter keeps pushing AC into the line, you can electrocute the lineworker repairing the pole. All grid-tied inverters must be UL-1741 / IEC 62116 certified to detect grid loss within 2 seconds and shut down automatically.



Worked Example: A 1.5 kWh/day Off-Grid Cabin

Bringing the five modules together — design a complete system for a small off-grid cabin in Tigray, Ethiopia (mean daily irradiance $\approx 5.8 \text{ kWh/m}^2/\text{day}$, lowest month 5.0).

Stage	Calculation	Result
Daily load	$5 \times \text{LED } 10 \text{ W} \times 5 \text{ h} + 1 \times \text{fridge } 60 \text{ W} \times 24 \text{ h} + 1 \times \text{laptop } 50 \text{ W} \times 6 \text{ h}$	$\approx 2\,240 \text{ Wh/day}$
Array size	$2\,240 \div 5.0 \text{ (worst month)} \div 0.75 \text{ (losses)}$	$\approx 600 \text{ W} \rightarrow 2 \times 320 \text{ W}$
Battery bank	$2\,240 \times 2 \text{ days} \div 0.80 \text{ DoD} \div 0.92 \eta_{\text{inv}}$	$\approx 6\,100 \text{ Wh} \rightarrow 480 \text{ Ah Li}$
Charge controller	$640 \text{ W} \div 12.8 \text{ V} \times 1.25$	$\approx 63 \text{ A} \rightarrow 70 \text{ A MPPT}$
Inverter	Continuous 600 W steady, fridge surge 4×	1 500 W pure sine

✓ TIP Order of design matters

Always size in this order: load → array → battery → controller → inverter → wiring. Every later stage depends on numbers from the earlier ones — never start by buying a panel because it was on sale.



Self-Check Quiz

Five short questions to test understanding before moving on. Answers follow each question in italics — try to answer first, then read the explanation.

1. Why is an MPPT controller often more efficient than a PWM controller?

An MPPT controller is a DC-DC converter — it can pull the panel down to its Maximum Power Point (V_{mp}) and step that voltage down to the battery's level while keeping the power product constant. A PWM controller simply pulls the panel voltage all the way down to the battery, wasting any voltage above that level.

2. A 12 V battery bank must supply 1200 Wh/day at 50% DoD with 1 day autonomy. What is the minimum Ah size?

$Wh \div DoD \div V \rightarrow 1200 \div 0.5 \div 12 = 200 \text{ Ah}$. With 2 days autonomy it would double to 400 Ah.

3. Why does a panel's Voc increase in cold weather?

Semiconductor band-gap energy rises slightly as temperature drops, so each cell produces a higher open-circuit voltage. Always design controller input voltage for the coldest expected ambient — not the nameplate Voc at 25 °C.

4. What does 'anti-islanding' mean for a grid-tied inverter?

It is the automatic disconnect that prevents the inverter from feeding live voltage into a dead utility line — protecting line workers and preventing equipment damage when the grid recovers out-of-phase.

5. A modified-sine inverter is powering a microwave. What can you expect?

Possible buzzing, hotter transformer, reduced cooking power, and shortened component life. Microwaves, audio gear, variable-speed motors and medical equipment all require pure sine output.

i INFO Going further

If you scored 4 / 5 or higher, you are ready to size your own pilot system. Visit ayetechub.com for the full 30-day AI & Engineering course, downloadable PDFs, and our Tigray-based installation services.



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About the author

Awet G. Nway is a Robotics, Automation and MEP Engineering specialist with hands-on experience in industrial automation, machine foundation supervision, production-and-maintenance team leadership, and renewable-energy systems. He is the founder of AYE Tech Hub, an engineering education and services platform serving learners and businesses across Ethiopia and beyond.

Contact & resources

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